

Multimedia Computing

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Content

- Arithmetic Coding Algorithm
- Scaling and incremental coding
- Lossy Compression Algorithms
- Advantage and disadvantage
- MPEG1
- MP3
- MP4

Arithmetic Coding

- ▶ Arithmetic coding is a more modern coding method that usually out-performs Huffman coding.
- ▶ Huffman coding assigns each symbol a codeword which has an integral bit length. Arithmetic coding can treat the whole message as one unit.
- ▶ A message is represented by a half-open interval $[a, b)$ where a and b are real numbers between 0 and 1. Initially, the interval is $[0, 1)$. When the message becomes longer, the length of the interval shortens and the number of bits needed to represent the interval increases.

► ALGORITHM 7.5 Arithmetic Coding Encoder

BEGIN

```
    low = 0.0;    high = 1.0;    range = 1.0;
```

```
    while (symbol != terminator)
```

```
    {
```

```
        get (symbol);
```

```
        low = low + range * Range_low(symbol);
```

```
        high = low + range * Range_high(symbol);
```

```
        range = high - low;
```

```
    }
```

```
    output a code so that low ≤ code < high;
```

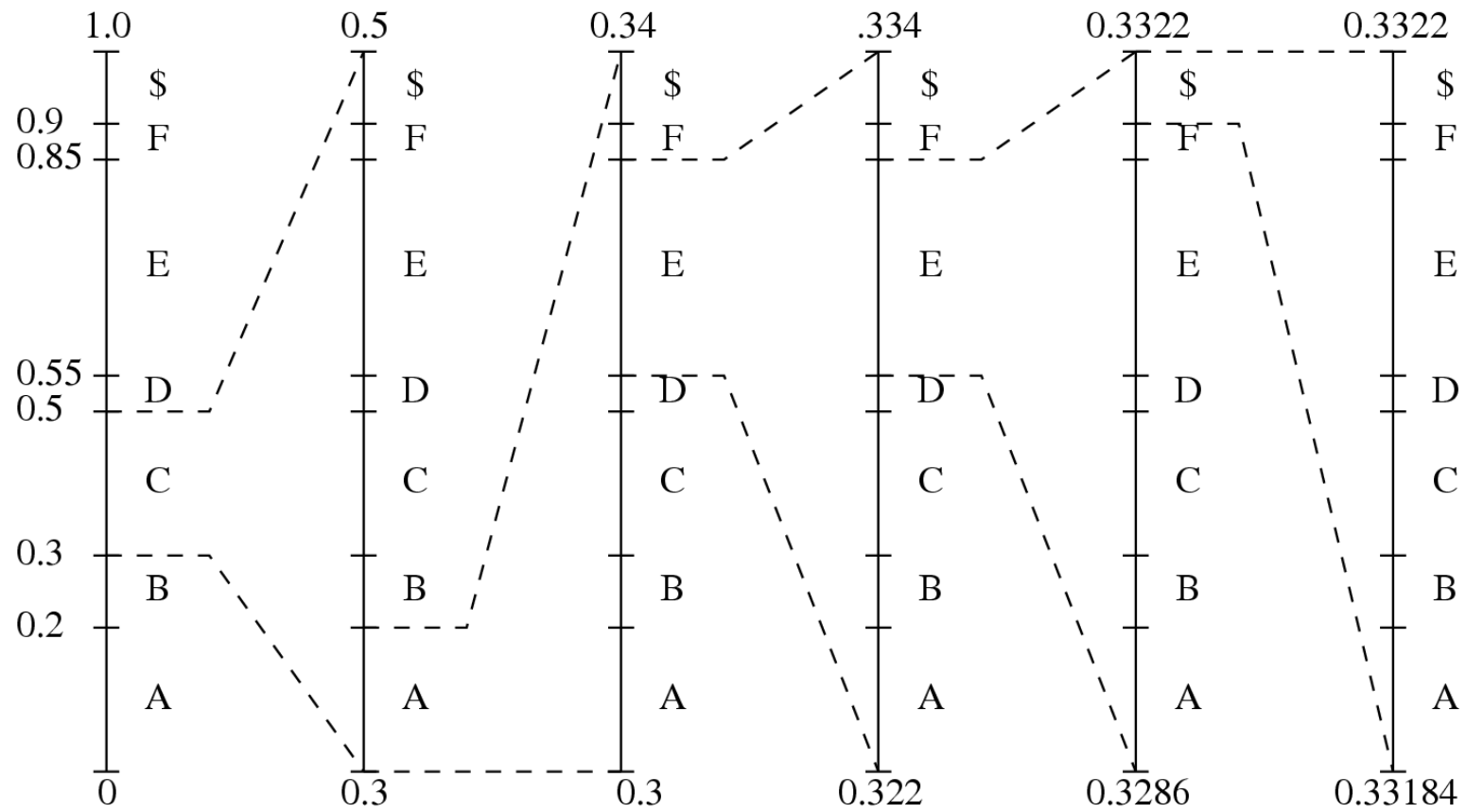
END

► Example: Encoding in Arithmetic Coding

Symbol	Probability	Range
A	0.2	[0, 0.2)
B	0.1	[0.2, 0.3)
C	0.2	[0.3, 0.5)
D	0.05	[0.5, 0.55)
E	0.3	[0.55, 0.85)
F	0.05	[0.85, 0.9)
G	0.1	[0.9, 1.0)

(a) Probability distribution of symbols.

► Fig. 7.8: Arithmetic Coding: Encode Symbols “CAEE\$”



► Fig. 7.8(b) Graphical display of shrinking ranges.

► Example: Encoding in Arithmetic Coding

Symbol	Low	High	Range
	0	1.0	1.0
C	0.3	0.5	0.2
A	0.30	0.34	0.04
E	0.322	0.334	0.012
E	0.3286	0.3322	0.0036
\$	0.33184	0.33220	0.00036

► (c) New *low*, *high*, and *range* generated.

► Fig. 7.8 (cont'd): Arithmetic Coding: Encode Symbols “CAEE\$”

► PROCEDURE 7.2 Generating Codeword for Encoder

```
BEGIN
    code = 0;
    k = 1;
    while (value(code) < low)
    {
        assign 1 to the kth binary fraction bit
        if (value(code) > high)
            replace the kth bit by 0
        k = k + 1;
    }
END
```

The final step in Arithmetic encoding calls for the generation of a number that falls within the range $[low, high)$. The above algorithm will ensure that the shortest binary codeword is found.

► ALGORITHM 7.6 Arithmetic Coding Decoder

BEGIN

 get binary code and convert to

 decimal value = value(code);

 Do

 {

 find a symbol s so that

$\text{Range_low}(s) \leq \text{value} < \text{Range_high}(s)$;

 output s ;

$\text{low} = \text{Rang_low}(s)$;

$\text{high} = \text{Range_high}(s)$;

$\text{range} = \text{high} - \text{low}$;

$\text{value} = [\text{value} - \text{low}] / \text{range}$;

 }

 Until symbol s is a terminator

END

► Table 7.5 Arithmetic coding: decode symbols “CAEE\$”

Value	Output Symbol	Low	High	Range
0.33203125	C	0.3	0.5	0.2
0.16015625	A	0.0	0.2	0.2
0.80078125	E	0.55	0.85	0.3
0.8359375	E	0.55	0.85	0.3
0.953125	\$	0.9	1.0	0.1

Lossy Compression

- The compressed data is not the same as the original data but a close approximation of it.
- Yields a much higher compression ratio than that of loss less compression

Frequency Masking

- Lossy audio data compression methods, such as MPEG/Audio encoding, remove some sounds which are masked anyway
- The general situation in regard to masking is as follows:
 1. A lower tone can effectively mask (make us unable to hear) a higher tone
 2. The reverse is not true – a higher tone does not mask a lower tone well
 3. The greater the power in the masking tone, the wider is its influence – the broader the range of frequencies it can mask.
 4. As a consequence, if two tones are widely separated in frequency then little masking occurs

Threshold of Hearing

- A plot of the threshold of human hearing for a pure tone

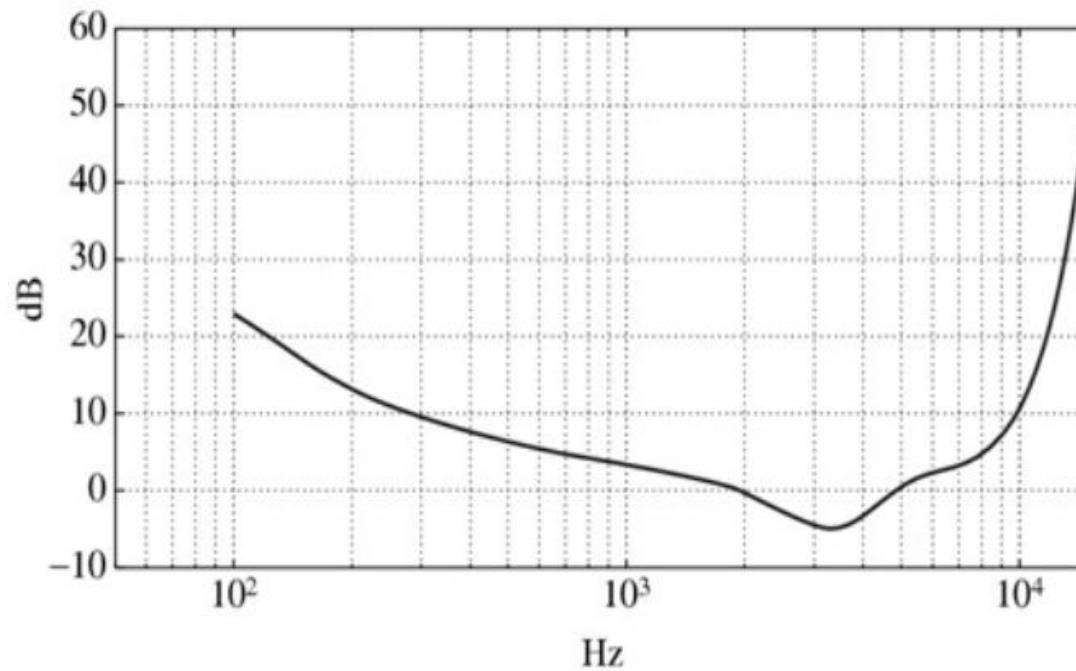


Fig. 14.2: Threshold of human hearing, for pure tones

Threshold of Hearing

- The threshold of hearing curve: if a sound is above the dB level shown then the sound is audible
- Turning up a tone so that it equals or surpasses the curve means that we can then distinguish the sound
- An approximate formula exists for this curve:

$$\text{Threshold}(f) = 3.64(f / 1000)^{-0.8} - 6.5 e^{-0.6(f/1000-3.3)^2} + 10^{-3}(f / 1000)^4$$

(14.1)

– The threshold units are dB; the frequency for the origin (0,0) in formula (14.1) is 2,000 Hz: $\text{Threshold}(\hat{f}) = 0$ at $\hat{f}=2$ kHz

Why Audio Compression is Needed

- ▶ to reduce the span between the softest and loudest parts of an audio signal
- ▶ Data rate = sampling rate * quantization bits * channels (+ control information)
- ▶ For example (digital audio):
 - ▶ 44100 Hz; 16 bits; 2 channels
 - ▶ generates about 1.4M of data per second; 84M per minute; 5G per hour

Audio compression format

- ▶ An **audio coding format** (or **audio compression format**) is a content representation format for storage or transmission of digital audio (such as in digital television, digital radio and in audio and video files). Examples of audio coding formats include MP3, AAC, Vorbis, FLAC, and Opus.
- ▶ A specific software or hardware implementation capable of audio compression and decompression to/from a specific audio coding format is called an *audio codec*;

Lossless audio coding

- ▶ Lossless audio coding formats such as FLAC and Apple Lossless are sometimes available, though at the cost of larger files.

Audio compression Lossy

- ▶ The most widely used audio coding formats are MP3 and Advanced Audio Coding (AAC), both of which are lossy formats based on modified discrete cosine transform (MDCT) and perceptual coding algorithms
- ▶ Loss of some original audio data, permanently
- ▶ Lower audio quality, noticeable in certain scenarios
- ▶ Not suitable for high-end audio applications (music production, audiophiles)

History of MPEG Audio - MP3

- ▶ **First psychoacoustic masking code** was proposed in 1979 in AT&T - Bell Labs, Murray Hill.
- ▶ MP3 based on OCF (optimum coding in frequency domain) and PTFM (Perceptual transform coding)
- ▶ MPEG-1 Audio Layer III - public release 1993
- ▶ MPEG-2 Audio III - public release 1995

MPEG-1 Audio

- ▶ Lossy compression of audio
- ▶ In late 1980's ISO's MPEG group started to standardize
 - ▶ TV broadcasting
 - ▶ Use of Audio on CD-ROM (later DVD)
- ▶ MPEG-1 Audio - 1992
- ▶ MPEG-2 Audio - 1994
- ▶ MPEG-1 Audio Layer I, II, III

MPEG-1 Audio Encoding

▶ Characteristics

- ▶ Precision 16 bits
- ▶ Sampling frequency: 32KHz, 44.1 KHz, 48 KHz
- ▶ 3 compression layers: Layer 1, Layer 2, Layer 3 (MP3)
 - ▶ Layer 3: 32-320 kbps, target 64 kbps
 - ▶ Layer 2: 32-384 kbps, target 128 kbps
 - ▶ Layer 1: 32-448 kbps, target 192 kbps

MPEG-1 Audio Layer II

- ▶ Called **MP2**
- ▶ Dominant standard for **audio broadcasting**
 - ▶ **DAB digital radio and DVB digital television**
- ▶ Came out of MUSICAM codecs with bit rates 64-196 kbps
 - ▶ MUSICAM audio coding - basis for MPEG-1 and MPEG-2 audio
- ▶ Sampling rates: 32, 44.1, 48 kHz
- ▶ Bit rates: 32, 48, 56, 64, 80, 96, ... 384 kbps
- ▶ Format: mono, stereo, dual channel, ...
 - ▶ MP2 - sub-band audio encoder in **time domain**

MPEG-1 Audio Layer III

- ▶ MPEG-1 Layer III is called **MP3** format
 - ▶ Popular for Internet applications
 - ▶ Goal to compress to **128 kbps**, but can be compressed to higher or lower resulting quality
 - ▶ Utilization of **psychoacoustics**
 - ▶ Scientific study of sound perception .

MP3 (MPEG Audio Layer III)

- ▶ The Motion Pictures Experts Group (MPEG), a group that works on standards for both audio and video systems. MPEG is a working group that was formed from the existing JPEG (Joint Photographic Experts Group) standard for bit reduction of still pictures. Its goal is to devise a suitable encoding scheme for transmitting moving pictures and sound over various broadcast links and recording them in standard digital storage media. Since moving pictures are often accompanied by sound, MPEG also defined a standard for encoding audio information. In 1992, a standard for audio and video encoding was devised known as MPEG-1.

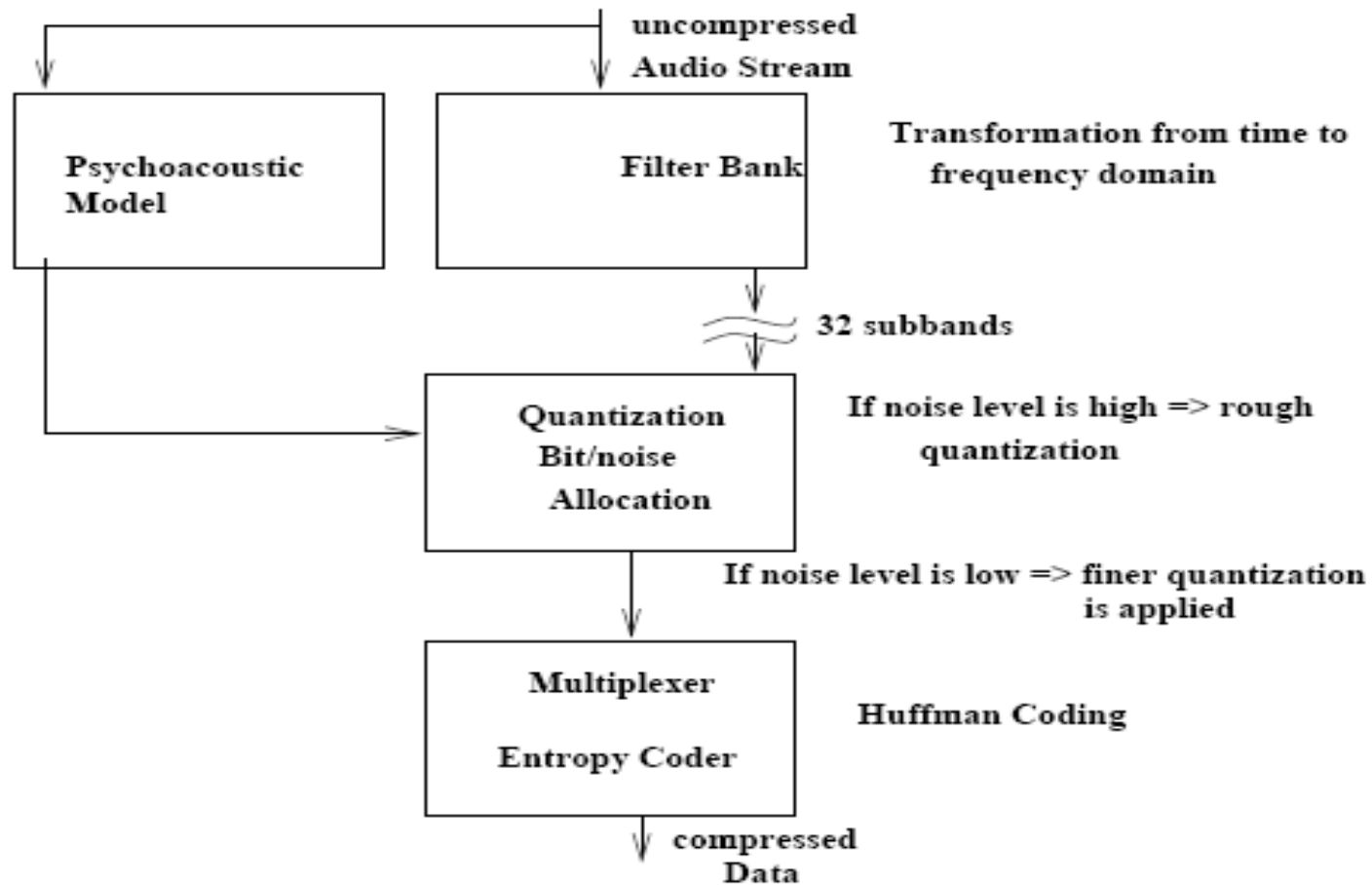
MP3 (MPEG Audio Layer III)

- ▶ MP3 stands for MPEG-1 Layer 3. Layers represent a family of coding algorithms. Layer 1 has the lowest complexity compared to the other layers. Layer 2 requires a more complex encoder and decoder and is directed more towards different applications. Thus, Layer 2 is more able than Layer 1 to remove the redundancy in the signal that takes up space. Layer 3 is again more complex and further reduces redundancy and relevance from the data. . The layers are backwards-compatible, which means that any software or hardware capable of decoding Layer 3 audio should also be able to decode Layers 1 and 2.

MP3 (MPEG Audio Layer III)

- ▶ MPEG-1 Layer 3 is defined as an open standard, so the specification is available to anyone interested in implementing the standard. Since no single company owns the standard, public example source code is available and the format is well defined. Also, driven by the demand for professional use, many MP3 encoders and decoders are readily available for use. This simplifies and accelerates the adoption of MP3 technology. Additionally, the widespread increase in usage of computers fast enough to decode audio, as well as increasingly fast and widely available Internet access, has contributed to the embracing of MP3 music format.

MPEG Audio Encoding Steps



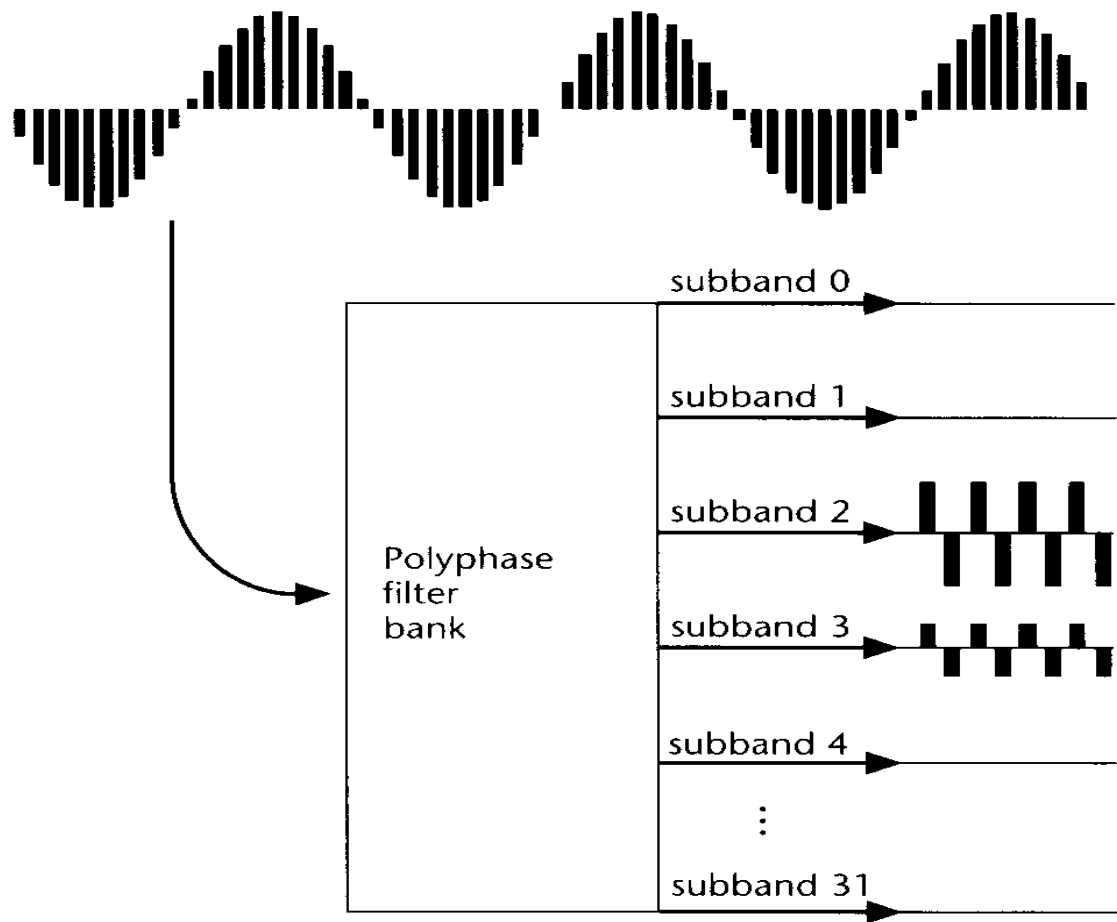


Figure 5. Aliasing: Pure sinusoid input can produce nonzero output for two subbands.

Subband outputs:
8x32 samples; both subbands 3 and 4
have significant output values

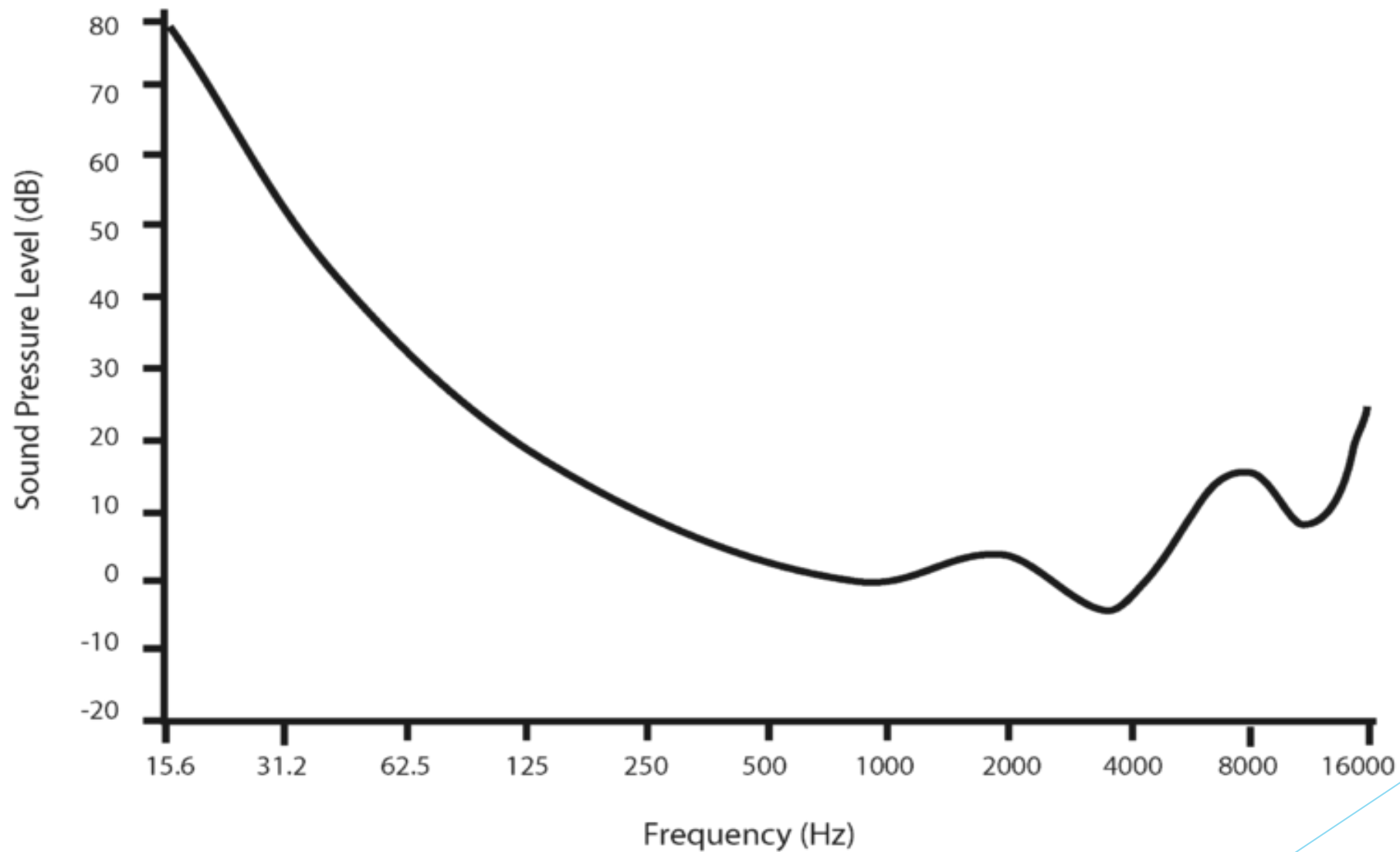
MPEG Audio Filter Bank

- ▶ **Filter bank** divides input into multiple sub-bands (32 equal frequency sub-bands)

MPEG Audio Psycho-acoustic Model

- ▶ Compresses by **removing acoustically irrelevant parts** of audio signals
- ▶ Takes advantage of **human auditory systems inability to hear quantization noise** under **auditory masking**
 - ▶ **Auditory masking**: occurs when ever the presence of a strong audio signal makes a temporal or spectral neighborhood of weaker audio signals imperceptible.

Perceived Human Hearing



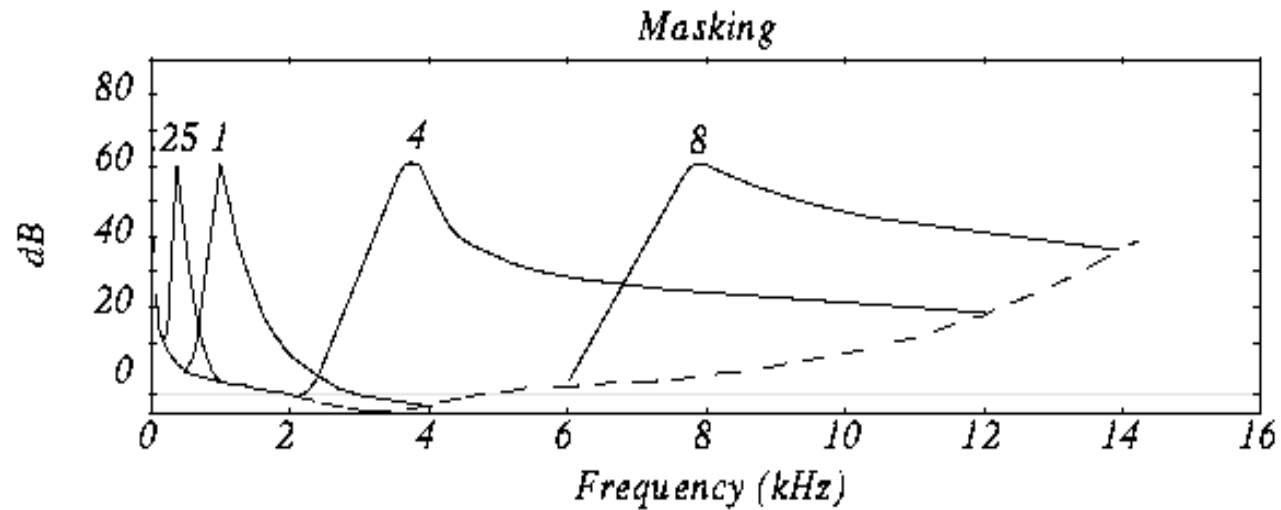
Loudness and Pitch

(Review on Psychoacoustic Effects)

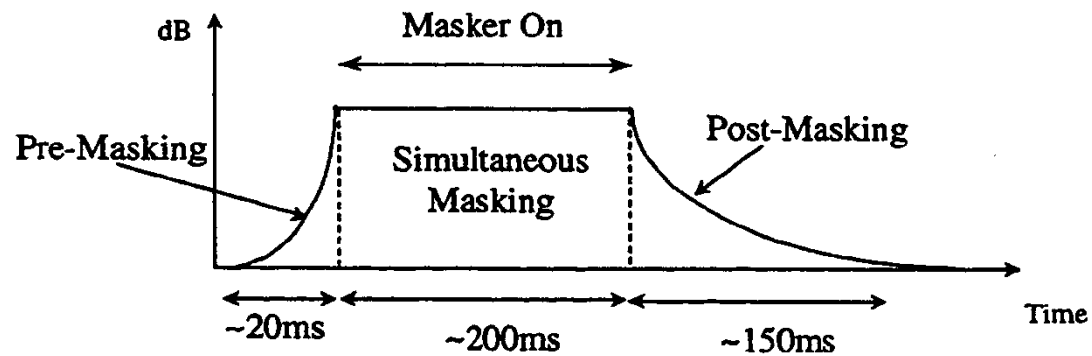
- ▶ More sensitive to loudness at **mid frequencies** than at other frequencies
 - ▶ intermediate frequencies at [500hz, 5000hz]
 - ▶ Human hearing frequencies at [20hz,20000hz]
- ▶ Perceived **loudness** of a sound changes based on **frequency** of that sound
 - ▶ basilar membrane reacts more to intermediate frequencies than other frequencies

Masking Effects

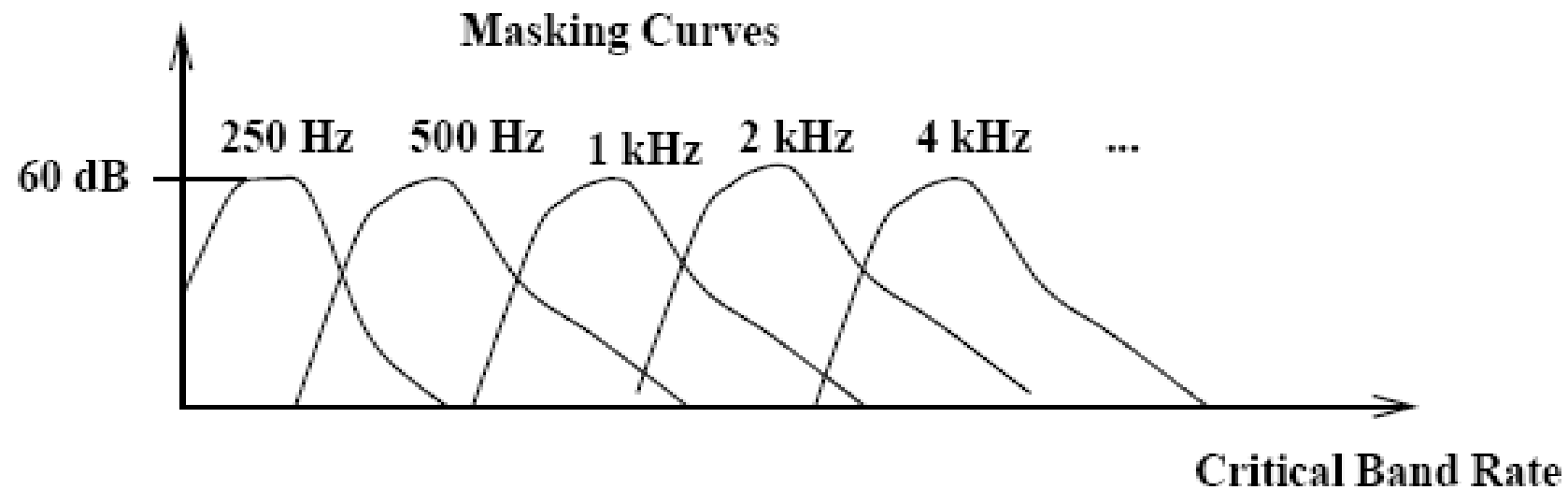
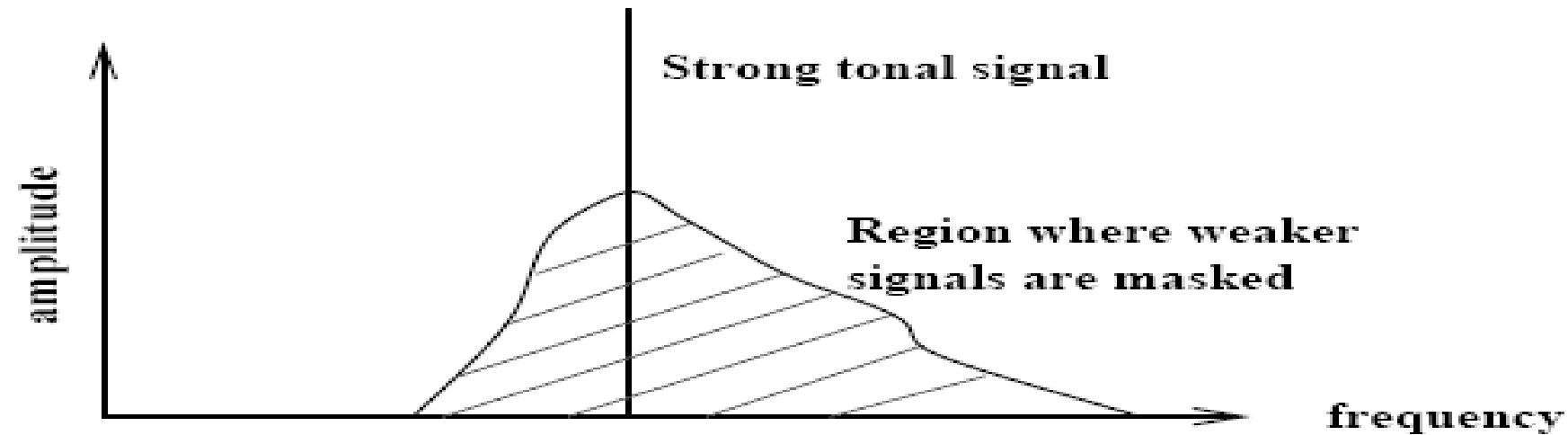
(Review of Psychoacoustic Effects)



Frequency masking



Temporal masking



MPEG/audio divides audio signal into frequency sub-bands that approximate critical bands. Then we quantize each sub-band according to the audibility of quantization noise within the band

MPEG Audio Bit Allocation

- ▶ This process determines number of code bits allocated to each sub-band based on information from the psycho-acoustic model
- ▶ Algorithm:
 1. **Compute mask-to-noise ratio:** $MNR = SNR - SMR$
 - ▶ Standard provides tables that give estimates for SNR resulting from quantizing to a given number of quantizer levels
 2. **Get MNR for each sub-band**
 3. **Search for sub-band with the lowest MNR**
 4. **Allocate code bits** to this sub-band.
 - ▶ If sub-band gets allocated more code bits than appropriate, look up new estimate of SNR and repeat step 1

Audio Quality

- ▶ **Bitrate**

- ▶ With too low bit rate, we get compression artifacts

- ▶ **Ringing**

- ▶ **Pre-echo** - sound is heard before it occurs. It is most noticeable in impulsive sounds from percussion instruments such as cymbals

- ▶ Occurs in transform-based audio compression algorithms

- ▶ Quality of **encoder and encoding parameters**

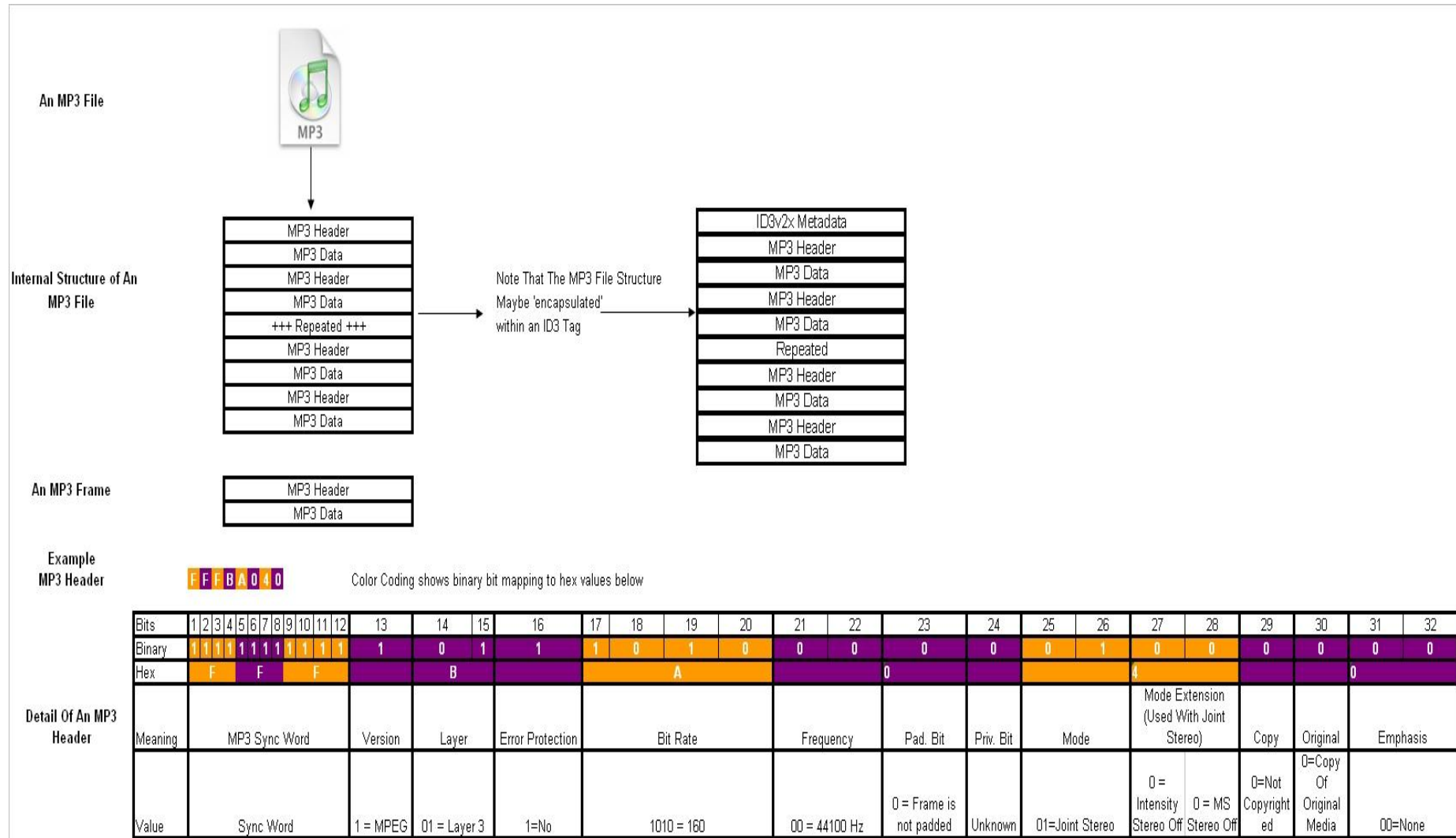
- ▶ Constant Bit rate encoding

- ▶ Variable Bit rate encoding

MP3 (MPEG Audio Layer III)

- ▶ A psychoacoustic model refers to an analysis of an audio signal that determines the amount of noise masking available based on frequency and loudness.
- ▶ Uses **psychoacoustic models** to eliminate sounds humans cannot hear.
- ▶ Reduces bitrate by focusing on the most important audio frequencies.
- ▶ High compression ratio. Suitable for music and voice compression with moderate to high quality

MP3 Audio Format



Source: <http://wiki.hydrogenaudio.org/images/e/ee/Mp3filestructure.jpg>

MPEG Audio Comments

- ▶ Precision of **16 bits per sample** is needed to get good SNR ratio
- ▶ Noise we are getting is **quantization noise** from the digitization process
- ▶ For each added bit, we get **6dB better SNR ratio**
- ▶ **Masking effect** means that we can raise the noise floor around a strong sound because the noise will be masked away
- ▶ Raising **noise floor** is the same as using **less bits** and using less bits is the same as **compression**

MPEG Audio - MP3

- ▶ 1997 - mp3.com - offering thousands of MP3s created by independent artists for free
- ▶ 1999 - Napster MP3 peer-to-peer file sharing
- ▶ Problem: copyright infringement
- ▶ Authorized services: Amazon.com, Rhapsody, Juno Records, ..

How Does Lossy Audio Compression Work?

- Perceptual coding: **Only keep audio data that humans can hear**
- Use of algorithms (e.g., MP3, AAC) to remove redundant or less important sounds
- Example of human hearing limitations (high-frequency sounds)

Advantages of Lossy Compression

- Small file sizes, saving storage space
- Faster transmission over the internet (streaming, downloading)
- Suitable for consumer-grade audio where perfect fidelity isn't necessary

Applications of Lossy Compression

- ▶ Music streaming (Spotify, Apple Music)
- ▶ Podcasts
- ▶ Audiobooks
- ▶ Online videos (YouTube)
- ▶ Video conferencing (Zoom, Skype)

Criteria for Good Standard

- ▶ Achieve desired outcome
 - ▶ Be comprehensible
 - ▶ Allow efficient implementation
 - ▶ Support competition
 - ▶ Give benchmark tests
 - ▶ Be supported by industry
 - ▶ Be good for end users
 - ▶
- ▶ Two models:
 - ▶ implement first, then standardize
 - ▶ standardize first, then implement

MPEG-4 Audio (Successor of MP3)

- ▶ **Advanced Audio Coding (AAC)** - now part of **MPEG-4 Audio**
- ▶ Inclusion of 48 full-bandwidth audio channels
- ▶ Default audio format for iPhone, iPad, Nintendo, PlayStation, Nokia, Android, BlackBerry
- ▶ Introduced 1997 as MPEG-2 Part 7
- ▶ In 1999 - updated and included in MPEG-4

MPEG-4 Audio

- ▶ Bit-rate 2-64kbps
- ▶ Scalable for variable rates
- ▶ MPEG-4 defines set of coders
 - ▶ **Parametric Coding Techniques**: low bit-rate 2-6kbps, 8kHz sampling frequency
 - ▶ **Code Excited Linear Prediction**: medium bit-rates 6-24 kbps, 8 and 16 kHz sampling rate
 - ▶ **Time Frequency Techniques**: high quality audio 16 kbps and higher bit-rates, sampling rate > 7 kHz

MPEG-4 Audio

(AAC's Improvements over MP3)

- ▶ Advanced Audio Coding in MPEG-4
- ▶ More **sample frequencies** (8-96 kHz)
- ▶ **Arbitrary bit rates** and variable frame length
- ▶ Higher efficiency and simpler filterbank
 - ▶ Uses pure MDCT (**modified discrete cosine transform**)
 - ▶ Used in Windows Media Audio

MPEG-4 Audio

- ▶ Variety of applications
 - ▶ General audio signals
 - ▶ Speech signals
 - ▶ Synthetic audio
 - ▶ Synthesized speech (structured audio)

MPEG-4 Audio

- ▶ Includes variety of audio coding technologies
 - ▶ **Lossy speech coding** (e.g., CELP)
 - ▶ CELP - code-excited linear prediction - speech coding
 - ▶ General audio coding (AAC)
 - ▶ Lossless audio coding
 - ▶ **Text-to-Speech interface**
 - ▶ **Structured Audio** (e.g., MIDI)

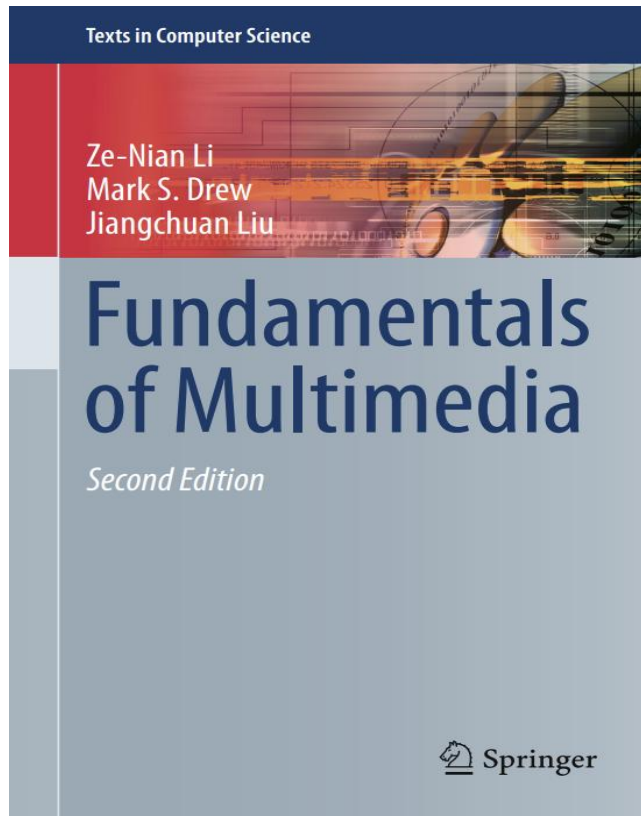
MPEG-4

- ▶ Called MP4 with Extension .mp4
- ▶ Multimedia **container format**
- ▶ Stores digital video and audio streams and allows streaming over Internet
- ▶ Container or wrapper format
 - ▶ **meta-file format** whose spec describes how different data elements and metadata in computer file

Conclusion

- ▶ MPEG Audio is an integral part of the MPEG standard to be considered together with video
- ▶ MPEG-4 Audio represents a major extension in terms of capabilities to MPEG-1 Audio

Reference



Chapter 8

Question



Thank you

